

Lecture-1

From the MERN acronym to how a browser thinks

MERN ACRONYM

URL → SERVER JOURNEY

BROWSER'S 3 ENGINES

COMPILER PHASES

M-E-R-N



MongoDB

NoSQL, document-based database
— stores data as flexible JSON-like
documents



Express.js

Backend framework — runs on top
of Node.js to structure servers &
APIs



React.js

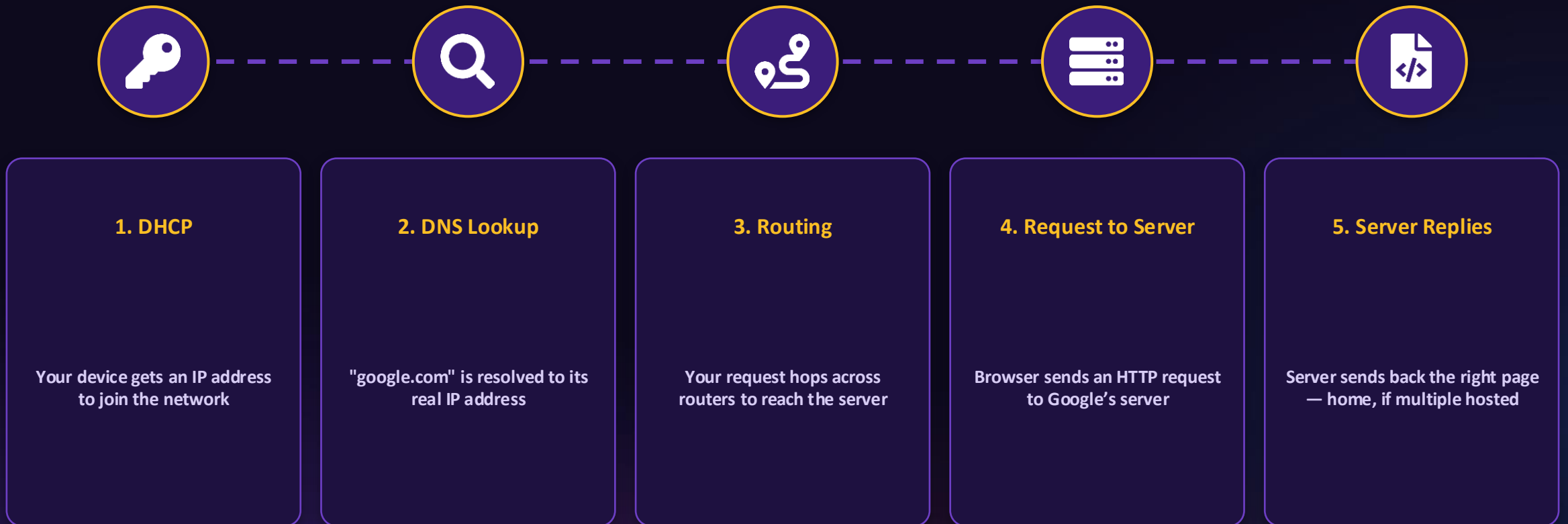
Frontend framework — builds the
interactive UI the user sees in the
browser



Node.js

JavaScript runtime environment —
lets JS run outside the browser, on
the server

What happens when you visit google.com



Five invisible steps happen before a single pixel appears on your screen.

Every browser can interpret 3 things



HTML Parser

Reads HTML tags and builds the page structure — the DOM tree



CSS Parser

Reads styles and paints colors, layout, fonts — the CSSOM

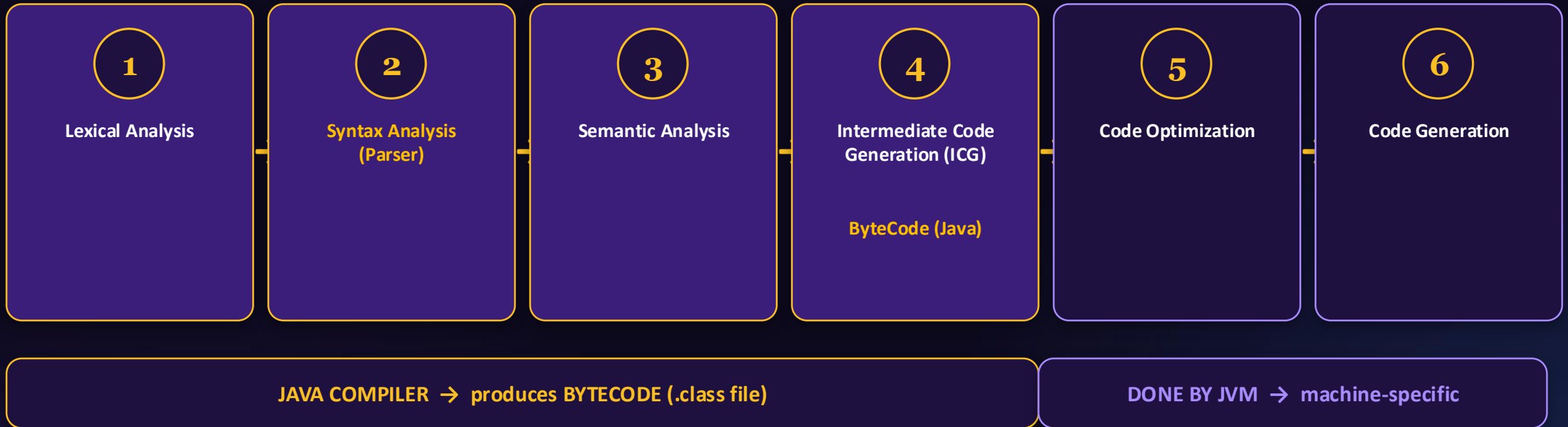


JS Engine

Executes JavaScript — adds interactivity & logic (e.g. V8)

All three combine to render the **final page you see on screen.**

Compiler phases — and where Java stops



Java stops at phase 4 so the SAME bytecode **runs on any machine** — the JVM finishes the last 2 phases per-platform.

This is exactly "Write Once, Run Anywhere."